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Apache mod_rewrite

the Swiss Army Knife of URL manipulation

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Intro

"The great thing about mod_rewrite is it gives you all the configurability and flexibility of Sendmail.
The downside to mod_rewrite is that it gives you all the configurability and flexibility of Sendmail."

-- Brian Behlendorf
Apache Group

Intro

"Despite the tons of examples and docs, mod_rewrite is voodoo.
Damned cool voodoo, but still voodoo. "

-- Brian Moore

Intro

But all this functionality and flexibility has its drawback: complexity.
So don't expect to understand this entire module in just one day.

docs/mod/mod_rewrite.html

Intro – agenda

- The main directives
  - RewriteRule
  - RewriteCond
- Comparing mod_rewrite to other solutions
- The directives
  - Main directives revisited
  - Other directives
- mod_rewrite solutions to common problems
- Questions

Intro – not covered

- Regular Expressions see:
  - perldoc perlre
  - "Mastering Regular Expressions" by Jeffrey Friedl

Directives – RewriteRule

Syntax: RewriteRule *Pattern Substitution*

Default: *None*

Context: server config, virtual host, directory, .htaccess

Override: *FileInfo*

- RewriteRule is the main directive in mod_rewrite
- RewriteRules are checked first
- RewriteRules are executed in the *URI translation* or *fixup* phase

Directives – RewriteCond

Syntax: RewriteCond *TestString CondPattern*

Default: *None*

Context: server config, virtual host, directory, .htaccess

Override: *FileInfo*

- RewriteCond is used to add conditions to RewriteRules
- One or more RewriteConds can precede a RewriteRule
- Multiple RewriteConds are joined by an implicit and unless [OR] is used
-

Imitation – Redirect

- The mod_alias version:

  Redirect /test/ http://example.com/test/

  RedirectPermanent /test2/ http://example.com/test2/

- The mod_rewrite version:

  RewriteEngine On

  RewriteRule ^/test/(.*) http://example.com/test/$1 [R]

  RewriteRule ^/test2/(.*) http://example.com/test2/$1 [R=301]

Imitation – ProxyPass

- The mod_proxy version

  ProxyPass /example/ http://example.com/

- The mod_rewrite version

  RewriteRule ^/example/(.*) http://example.com/$1 [P]

Imitation – UserDir

- The mod_userdir version

  UserDir /usr/web -> /usr/web/joe/one/two.html

  UserDir /home/*/web -> /home/joe/web/one/two.html

- The mod_rewrite version

  RewriteRule ^/~(.*)/(.*)$ /usr/web/$1/$2

  RewriteRule ^/~([a-z][a-z0-9]+)/(.*)$ /home/$1/web/$2

Imitation – mod_vhost_alias

The vhost_alias version:

```
VirtualDocumentRoot /vhosts/%0/docs/
```

- The mod_rewrite version1:

```
RewriteEngine On
RewriteRule  ^(.+)$       /vhosts/%{HTTP_HOST}/docs$1
```

- The mod_rewrite version2:

```
RewriteCond      %{HTTP_HOST}      ^(.*)$
RewriteRule      ^(.*)$  /vhosts/%1/docs$1
```

Imitation – trailing slash

- Why does accessing directories only work when I include the trailing "/"

  ServerName example.com

- The mod_rewrite solution:

  ```
  RewriteCond      %{REQUEST_FILENAME}      -d
  RewriteRule      ^(.+[^/])$       $1/      [R,L]
  ```

Directives – RewriteRule 2

- **RewriteRule** ***Pattern Substitution [Flags]***

[flags]:

- **'redirect|R [=code]'** (force redirect)
- **'proxy|P'** (force proxy)
- **'forbidden|F'** (force URL to be forbidden)
- **'gone|G'** (force URL to be gone)
- **'type|T=MIME-type'** (force MIME type)
- **'noescape|NE'** (no URI escaping of output)
- **'env|E=VAR:VAL'** (set environment variable)
- **'cookie|CO=NAME:VAL:domain[:lifetime[:path]]'** (set cookie)

Directives – RewriteRule 2b

- **RewriteRule** ***Pattern Substitution [Flags]***

[flags]:

  - **'last|L'** (last rule)
  - **'next|N'** (next round)
  - **'chain|C'** (chained with next rule)
  - **'skip|S=num'** (skip next rule(s))
  - **'passthrough|PT'** (pass through to next handler)
  - **'nosubreq|NS'** (used only if no internal sub-request)
  - **'qsappend|QSA'** (query string append)
  - **'nocase|NC'** (no case)

Directives – RewriteCond 2

- **Syntax: RewriteCond** ***TestString CondPattern [Flags]***
- *CondPattern* is matched like perl regular expression, + these:
  - Lexical (string) compares (< > =)
  - **'-d'** (is directory)
  - **'-f'** (is regular file)
  - **'-s'** (is regular file with size > 0)
  - **'-l'** (is symbolic link)

Directives – RewriteCond 2b

- **Syntax: RewriteCond *TestString CondPattern [Flags]***
  - '-F' (is existing file via subrequest)
  - '-U' (is existing URL via subrequest)

  All of these can be negated with !

- [Flags]
  - 'nocase|NC' (no case)
  - 'ornext|OR' (or next condition)

Directives – RewriteOptions

- **Syntax:** **RewriteOptions** ***Options***
- **Default:** RewriteOptions MaxRedirects=10
  - MaxRedirects
    - From 2.0.45 / 1.3.28
    - Example: RewriteRule ^(.*)$ /test/$1 [R]
  - Inherit
    - Inherit configuration from parent

Directives – Other 1

- **RewriteEngine** ***on*****|*****off*** (default off)
- **RewriteLog** ***file***
- **RewriteLogLevel** ***Level*** (default 0)
- **RewriteBase** ***URL-path*** (context: dir, .htaccess)

Directives – RewriteMap

Syntax: RewriteMap *MapName MapType*:*MapSource*

Default: not used per default

Context: server config, virtual host

- Defines a mapping function for key lookups
- MapType in one of:
  - int (toupper,tolower,escape,unescape)
  - prg (external program)
  - txt
  - rnd (like txt but makes random choice from | seperrated list)
  - dbm[=type] (one of gdbm, sdbm, ndbm or db – chek valid types with: ***bin/apxs -q AP_LIBS***)

Directives – RewriteMap 2

- **`${`** ***MapName*** **`:`** ***LookupKey*** **`}`**
- **`${`** ***MapName*** **`:`** ***LookupKey*** **`|`** ***DefaultValue*** **`}`**

Example (crude load balancer/service level selection):

```
RewriteMap  service      rnd:/some/where/map.txt
RewriteRule  ^/(.*)/(.*)$      http://${service:$1}/$2 [P]
```

/some/where/map.txt:

```
gold   big1|big2|big3|big4|big5|big6  # gold service
silver  small1                                  # min service
```

Examples 1

- No deep links / disallow external referers:

```
RewriteCond  %{HTTP_REFERER}    !=""
RewriteCond %{HTTP_REFERER} !^http://example.com/.*$ [NC]
RewriteRule .*    -     [F]
```

- Disallow UserAgent

```
RewriteCond %{HTTP_USER_AGENT} EvilBot.*
RewriteRule /nobot/.*     http://%{REMOTE_ADDR}/      [R=301]
```

Examples 2

- How to restructure homedirs to avoid getting too many directories

```
RewriteEngine on

RewriteRule     ^/~(([a-z])[a-z0-9]+)/(.*)
                /data/$2/$1/$3

or

RewriteMap    LC     int:tolower

RewriteRule     ^/~(([a-z])[a-z0-9]+)/(.*) /data/

    /data/${LC:$2}/${LC:$1}/$3        [NC]
```

Examples 3

- How to redirect a logged in user to a specific page:

```
RewriteEngine on
RewriteMap    user-to-page   txt:/path/to/map.user-to-page
RewriteCond   %{LA-U:REMOTE_USER}  ^(.+)$
RewriteRule   ^/$  ${user-to-page:%1|/notfound.html}

/path/to/map.user-to-page
user1 /some/where/user1.html
user2 /some/where/else/user2.html
...
```

Examples 4

- Webmail redirector

```
RewriteEngine on
RewriteCond   %{HTTP_HOST}  ^webmail\.(.*)\.dk$      [NC]
RewriteRule  ^/$       http://mail.dk/login.x?dom=%1    [R]
```

Examples 5

- Moving from index files to autoindex – a regular Redirect / download/index.htm /dir/ would loop

/download/.htaccess

```
RewriteEngine on
RewriteBase /download/
RewriteCond     %{REQUEST_FILENAME}     !-d
RewriteRule     ^(.*)/index\.html*      $1/             [R=301]
```

Examples 6

- Making sure dav users can't upload "evil" php

```
RewriteEngine on
RewriteCond     %{REQUEST_METHOD}   ^PUT$ [OR]
RewriteCond     %{REQUEST_METHOD}   ^MOVE$
RewriteRule     ^/dav/(.*)\.php$    /dav/$1.nophp
```

Examples 7

Problem: Ugly URLs.

http://www.server.com/cgi-bin/program.cgi?A=arg1&B=arg2

http://www.server.com/program/arg1/arg2

```
RewriteEngineOn
RewriteRule ^/program/([^/]+)/([^/]+) \
/serverroot/cgi-bin/program.cgi?A=$1&B=$2
```

(http://www.server.com/cgi-bin/program.cgi?A=$1&B=$2)

Questions



IRC:
irc.openprojects.net
nick: quasi

Latest version:
http://www.toftum.org/slides/ApacheCon/

mads@apache.org

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Click anywhere to return.

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